Combining Drugs to Treat Ovarian Cancer

Betsy

Ovarian Cancer Survivor, Florida

Betsy was diagnosed with stage IIC ovarian cancer in 2009. She underwent standard platinum-based chemotherapy, and her doctors gave her an all-clear. But a year later, scans revealed her cancer had returned. So, she found a phase I clinical trial for patients with ovarian cancer at the NCI-Designated Dana-Farber/Harvard Cancer Center in Massachusetts.

Ovarian cancer is the fifth leading cause of cancer death among women in the United States. “Approximately 70% of the women diagnosed with ovarian cancer will die from the disease,” explained Joyce Liu, M.D., a principal investigator of the trial. “It’s critical that we find more effective and better tolerated treatments to help women live better and live longer,” she said.

So, Joyce and her colleagues at Dana-Farber designed a trial combining the Food and Drug Administration-approved drug olaparib (Lynparza™) with the investigational drug cediranib. Olaparib inhibits an enzyme called PARP, and cediranbib inhibits the growth of blood vessels that tumors need to grow larger than about 1 mm in size.

Joyce received support from NCI in 2008, when she began developing the phase I trial, her first investigator-initiated trial as an early-career researcher. She drew on several NCI-funded preclinical studies that indicated these drugs might work much better together than either worked alone. “It’s an exciting possibility that we might make a tumor more vulnerable to a drug by adding a different class of drugs,” she reflected. “That’s the power of combination—having a true synergistic effect.”

Betsy enrolled in Joyce’s trial and spent the next several years commuting to Boston for treatment. Scans taken at 3- to 4-month intervals showed her tumors were shrinking, and, eventually, her doctors reported that they found no evidence of disease. Betsy and her family were elated.

Betsy continues to receive follow-up checks on the trial, which has moved successfully from a phase I and then phase II trial into an NCI-sponsored phase III trial to determine whether the combination regimen is more effective than olaparib or chemotherapy alone. Betsy hopes that scientists continue conducting research and developing drugs to help other patients with ovarian cancer.

Joyce agrees: “We need to understand what drives ovarian cancer at a biological level and find new ways to overcome resistance to platinum-based drugs. Combinations of therapies that build on our understanding of the various types of ovarian cancer will go a long way toward helping doctors better treat their patients.”

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